

Assignment Report

Title: **Compiler Principles**

The Twelfth Theory Homework

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[ABSTRACT]

After learning the lecture 12 of compiler principles[1], we got the homework at the end of the lecture.

There are some requirements for the homework:

- The file should be named as **“学号 _ 姓名 _ 第 i 次理论作业”**, where i is the number of the homework.
- The file should be in docx or **pdf** format.
- All the homework should be sent to the TA's email: **406648896@qq.com**
- The deadline for the homework is **24:00 on Sunday of the week** when the homework is assigned.

Keywords: Compiler Principles

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1 Exercise 12

1.1 Exercise 12.1

1.1.1 Problem

Given the following flow graph:

- Compute the dominators of all nodes.
- Find all back edges in the flow graph.
- Find all natural loops defined by each back edge.

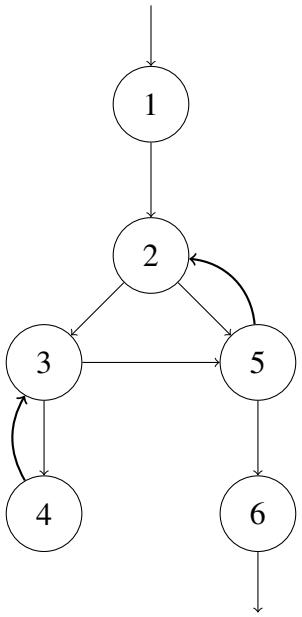


Figure 1.1 Flow graph for Exercise 12.1

1.1.2 Solution

1.1.2.1 Dominators of All Nodes

We compute the dominators using the standard dataflow algorithm. For each node n, $\mathrm{Dom}(n)$ is the set of nodes that dominate n.

- $\mathrm{Dom}(1) = \{1\}$ (node 1 is the entry and dominates itself)
- $\mathrm{Dom}(2) = \{1, 2\}$ (only 1 and 2 dominate 2)
- $\mathrm{Dom}(3) = \{1, 2, 3\}$ (1, 2, and 3 dominate 3)
- $\mathrm{Dom}(4) = \{1, 2, 3, 4\}$ (1, 2, 3, and 4 dominate 4)

- $\text{Dom}(5) = \{1, 2, 5\}$ (1, 2, and 5 dominate 5)
- $\text{Dom}(6) = \{1, 2, 5, 6\}$ (1, 2, 5, and 6 dominate 6)

1.1.2.2 Back Edges

A back edge is an edge (u, v) where v dominates u in the dominance tree (i.e., v comes before u in a depth-first spanning tree).

Back edges in this flow graph:

- Edge $(4, 3)$: Since $3 \in \text{Dom}(4)$, this is a **back edge**.
- Edge $(5, 2)$: Since $2 \in \text{Dom}(5)$, this is a **back edge**.

All other edges $(1, 2)$, $(2, 3)$, $(2, 5)$, $(3, 4)$, $(3, 5)$, and $(5, 6)$ are forward edges.

1.1.2.3 Natural Loops

For each back edge (u, v), the natural loop is the set of nodes that can reach u without going through v.

Natural loop for back edge $(4, 3)$**:**

$$\text{Loop}_{4\to3} = \{3, 4\}$$

This loop includes node 3 (the header) and node 4, since 4 reaches 3 directly via the back edge, and 3 only reaches 4 via the forward edge.

Natural loop for back edge $(5, 2)$**:**

$$\text{Loop}_{5\to2} = \{2, 3, 4, 5\}$$

This loop includes node 2 (the header), and all nodes that can reach 5 without passing through node 2. Because there is a forward edge $(3, 5)$, node 3 is also included, and then node 4 is included through the back edge $(4, 3)$. Thus the loop is $\{2, 3, 4, 5\}$.

Back Edge	Loop Header	Natural Loop
$(4, 3)$	3	$\{3, 4\}$
$(5, 2)$	2	$\{2, 3, 4, 5\}$

Table 1.1 Back edges and their natural loops

References

[1] JUN LI W. Lecture12. principles of compiler construction[Z]. 2026.